

EXAM 7 – SPRING 2012

1. (2.75 points)

Given the following information for accident year 2011 as of December 31, 2011:

- Accident year 2011 paid loss: \$700,000
- 2011 earned premium: \$3,000,000
- Initial expected loss ratio: 62.5%
- 12-24 month paid link ratio: 1.50
- 12-ultimate cumulative paid LDF: 2.50

a. (1.25 points)

Calculate accident year 2011 ultimate loss estimates as of December 31, 2011 using each of the following three methods:

- i. Chain ladder
- ii. Bornhuetter-Ferguson
- iii. Benktander

b. (1.5 points)

Determine the accident year 2011 incremental paid loss in 2012 that would result in the Benktander ultimate loss estimate being \$50,000 greater than the Bornhuetter-Ferguson ultimate loss estimate for accident year 2011, as of December 31, 2012. Assume all selected development factors remain the same.

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Question 1 Sample Answer

Solution 1

$$\begin{aligned}\text{a) AY ultimate chain ladder} &= \text{paid to date} \times \text{LDF}_{12-\text{ult}} \\ &= 700,000 \times 2.5 \\ &= 1,750,000\end{aligned}$$

$$\begin{aligned}\text{AY ultimate B-F} &= \text{Paid to date} + \text{expected ultimate} \times \% \text{ unpaid} \\ &= 700,000 + (3,000,000 \times .625) \times (1 - (1/2.5)) \\ &= 1,825,000\end{aligned}$$

$$\begin{aligned}\text{AY ultimate G-Benklander} &= \text{Paid to date} + \text{BF ultimate} \times \% \text{ unpaid} \\ &= 700,000 + 1,825,000 \times (1 - (1/2.5)) \\ &= 1,795,000\end{aligned}$$

$$24\text{-ultimate factor} = (2.5/1.5) = 1.667$$

$$\text{b) G-Benklander ultimate} = (700,000 + x) + U_{\text{BF}} \times (1 - (1/1.667))$$

$$\begin{aligned}\text{BF ultimate} &= (700,000 + x) + (3,000,000 \times .625) \times (1 - (1/1.667)) \\ &= 700,000 + x + 750,000 \\ &= 1,450,000 + x\end{aligned}$$

$$\begin{aligned}\therefore \text{G-Benklander ultimate} &= (700,000 + x) + (1,450,000 + x) (1 - (1/1.667)) \\ &= 700,000 + x + (1,450,000 + x) (0.4) \\ &= 700,000 + x + 580,000 + 0.4x \\ &= 1,280,000 + 1.4x\end{aligned}$$

$$\text{Benklander ultimate} = \text{BF ultimate} + 50,000$$

$$\begin{aligned}1,280,000 + 1.4x &= 1,450,000 + x + 50,000 \\ &= 1,500,000 + x\end{aligned}$$

$$0.4x = 220,000$$

$$x = 550,000$$

Sample 2

$$\begin{aligned}\text{a) Chain ladder} &700k \times 2.5 = 1.75M \\ \text{BF} &700k + (3000k) (0.625) (1 - (1/2.5)) = 1.825M \\ \text{Benklander:}\end{aligned}$$

$$p_k = 1/2.5 = 0.4$$

$$q_k = 1 - 0.4 = 0.6$$

$$(0.6^2) (3000k) (0.625) + (1 - 0.6^2) (1.75M) \\ = 675 + 1120 = 1795k$$

b) 24-ult LDF = $2.5/1.5 = 5/3$

$$p_{24} = 0.6 \quad q_{24} = 0.4$$

$$Ult_{Benk} - Ult_{BF} = Resv_{Benk} - Resv_{BF} = q_K U_{BF} - q_K U_0 = 50,000$$

$$= (0.4) [x + 0.4 (3000k) (0.625)] - (0.4) (3000k) (0.625) = 50k$$

$$x - (0.6) (3000k) (0.625) = 125k$$

$$x = 1250k$$

$$2012 \text{ incremental paid loss} \quad 1250k - 700k = 550k$$

Examiner Comment

The a. part of this question was fairly straightforward. Common errors were arithmetic errors and errors in the formulas for each method.

The b. part involved setup of both the Bornheutter-Ferguson and the Benktander methods and solving a system of equations to determine the incremental paid loss. Common errors were in calculating q_k , errors in recognizing that part b is a different time (12/31/2012) period than part a (12/31/2011), errors in the setup of the problem, and other calculation errors.